

Differential Gene Expression and Sample Clustering in a subtype of Acute Lymphoblastic Leukemia

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July 13, 2025

1 Background

In this report, our objective is to identify differentially expressed (DE) genes between patients diagnosed with acute lymphoblastic leukemia (ALL), specifically the Hyperdiploid>50 subgroup, and healthy individuals (a control group). In addition we explore potential subgroups on the subset of ALL Hyperdiploid>50 cases, by means of a cluster analysis.

The dataset is comprised of 45 samples of gene expression data, measured with Affymetrix Genechips. From these, 26 samples are from individuals with ALL (in the Hyperdiploid>50 subgroup), while 19 are of healthy individuals. The chips measure the gene expression of 12625 probe sets, among which there are 67 control probe sets. The data comes from the following study: [10].

We perform our analysis with the statistical software R, and employ packages developed by the Bioconductor project.

2 RMA

To obtain normalised values for the expression of each probe set, we use the summary statistic called Robust Multi-array Analysis (RMA), as described in [4]. As its name suggests, it is robust (so it down-weights outliers, making the model more precise) and considers each chip in the context of the whole set of arrays. It only uses the perfect-match intensity (PM) of the chips, and consists of three steps:

Background correction: On each chip, the measured intensity I is modelled as a sum of random variables: $I = S + B$, where B is the background (non-biological signal), and S is the true signal (originated from biological expression). RMA estimates S under the premise that B has a normal distribution, and S an exponential one.

Quantile normalisation: The chips may also capture a different total intensity from one another. To force consistency, consider M the matrix of log-transformed PM intensity, where columns represent probes, and row chips. Suppose $M \in \mathbb{R}_{n \times m}$. Then,

1. We get a bijection $R : \{1, \dots, n\} \times \{1, \dots, m\} \rightarrow \{1, \dots, n\} \times \{1, \dots, m\} : (i, j) \mapsto (i, r)$ such that $M_{i,j}$ is the r -th lowest value in the i -th row of M . In other words, we rank probe intensities by ascending order in each chip.
2. Then, the normalised values are given by the matrix M' , defined so that for any pair (i, j) ,

$$M'_{i,j} = \frac{1}{m} \sum_{x=1}^m M_{R^{-1}(x,r)}$$

where r is such that $R(i, j) = (i, r)$. That is, we replace each probe's intensity with the average of values with same rank across chips.

Estimation: We want to estimate the expression of each probe set. For this, we assume a linear additive model on $\log_2$ scale of the form

$$\log_2(PM_{i,j}) = a_i + b_j + e_{ij}$$

where we have¹ $PM_{i,j} := M'_{i,j}$, a_i is the chip effect ($\log_2$ of the gene's expression in the chip), b_j the probe effect (bias that the probe may have) and e_{ij} an error term.

¹For a probe set g , we only take pairs i, j corresponding to a probe of g

Ultimately, we are interested in the estimation of the a_i . For this we use the affyPLM package [3], which employs robust regression², a generalisation of linear regression. In particular, we minimise an objective function of the form

$$\min\left\{\sum_{i,j} w_{i,j} * (\log_2(PM_{i,j}) - a_i - b_j)\right\}$$

where $w_{i,j} \in [0, 1]$ are weights given by a fixed function (in our case, we use Huber's weight function).

Robust regression also allows us to do quality diagnostics, as we detail in the next section.

3 Quality control (QC)

We would like to verify that the chips do not present any influential defect. To do so, we try to identify those whose data is more extreme relative to the whole sample. We use the package affyPLM and follow the process described in section 3.5.1 of [3]:

For all chips, we assign to each probe its respective weight given by robust regression. We represent this assignment in a pseudo-image, a colouring of each probe of the chip.

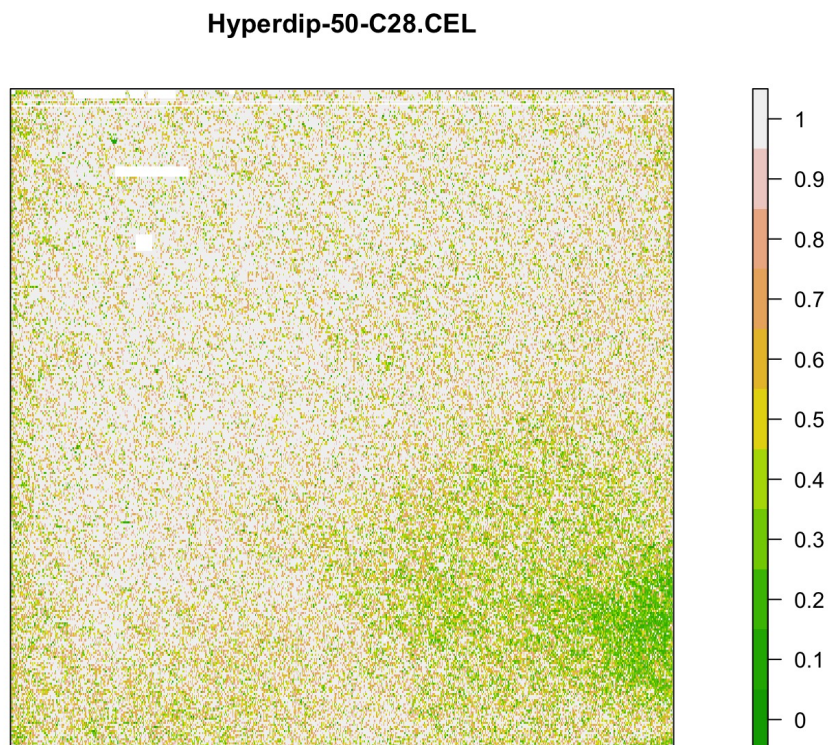


Figure 1: Weight pseudo-image of a chip. For the following ones, the legend is the same

In addition, we also compute the normalised unscaled standard error (NUSE): fix a probe set g and a chip i . With RMA, we obtain the chip effect as an estimate $\hat{\mu}_{i,g}$. Then, its NUSE is its standard error $SE(\hat{\mu}_{i,g})$, normalised with the median of standard errors across chips:

$$NUSE(g, i) := \frac{SE(\mu_{i,g})}{med_x(SE(\mu_{x,g}))}$$

²Another option would be to use median polish, which is a cheaper method. As we want to do quality control, we choose robust regression.

3.1 Interpretation of chip quality

We hypothesise that the chips in the left column below (Figure 2) may have quality issues, due to the artifacts in their pseudo-images:

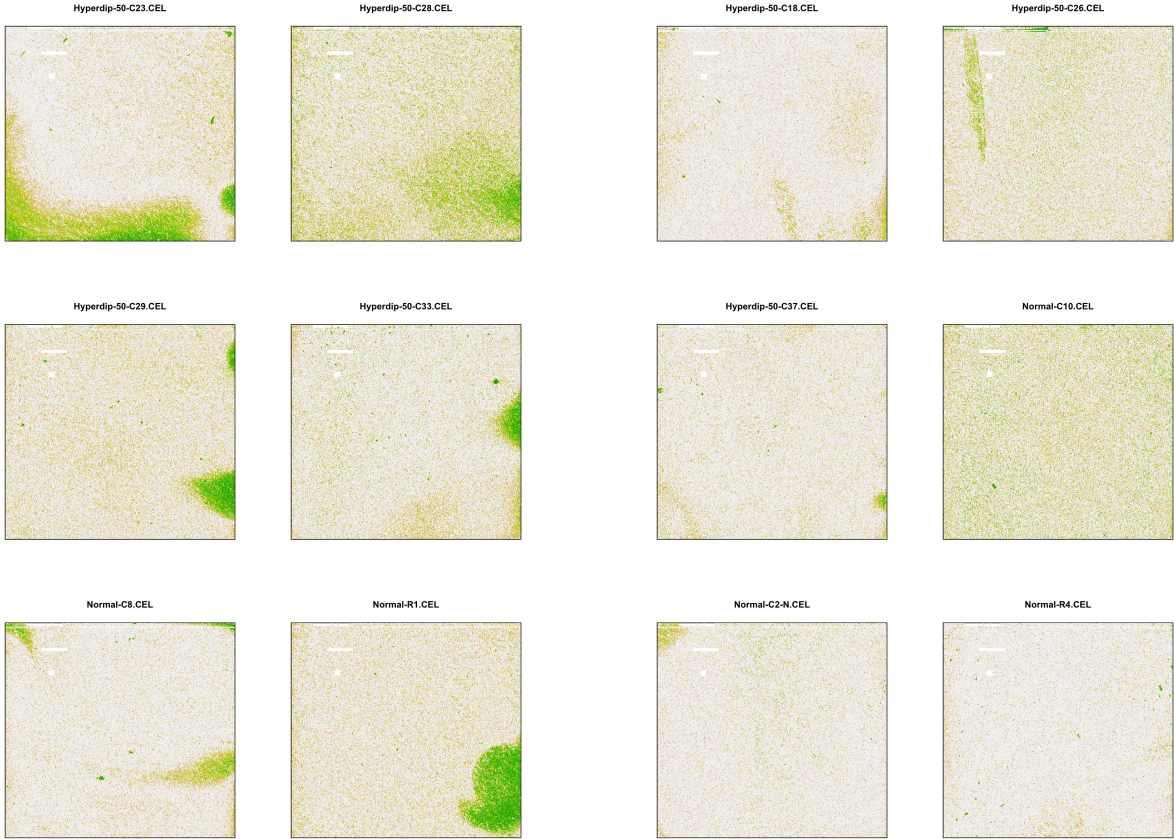


Figure 2: Weights pseudo-images that we determine to present artifacts

Figure 3: Pseudo-images of other chips as comparison.

When we study their NUSE, the right uppermost chip in Figure 2 (Hyperdip-50-C28) stands out: it has a highly deviated median, over 1.05, and high IQR. Figure 4 shows this visually:

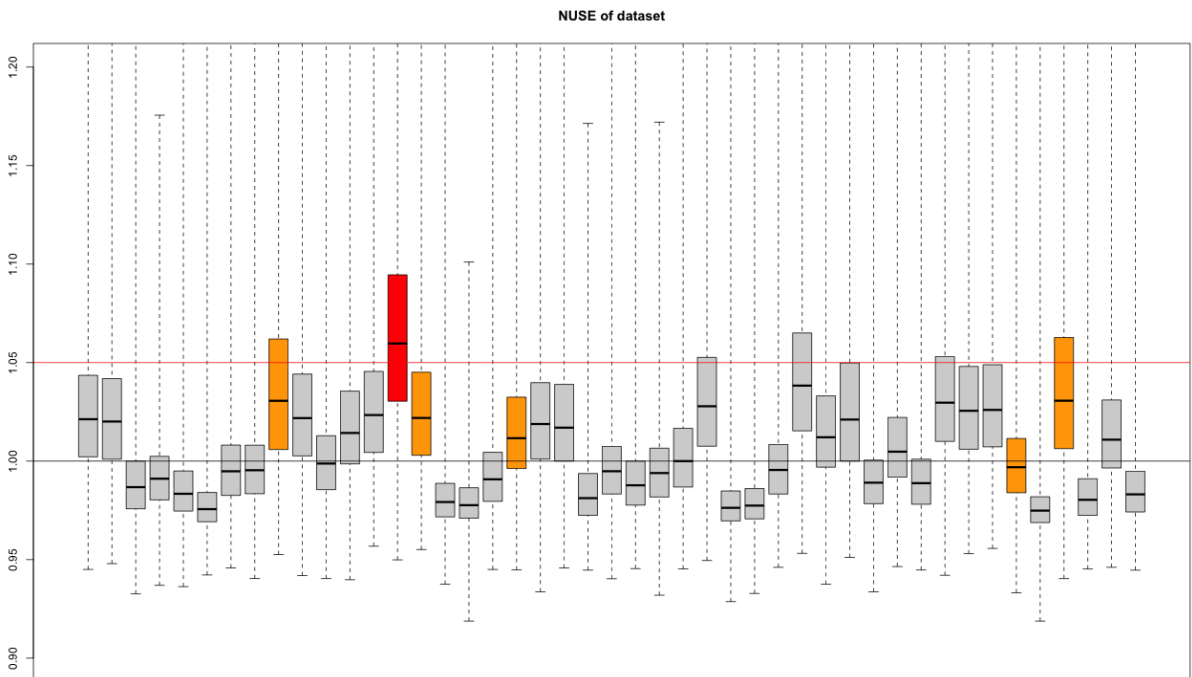


Figure 4: NUSE boxplots of chips. The orange ones correspond to the chips of Figure 2, and the red one to the discarded one

Therefore, we consider this chip to be of bad quality. We do not use it for the rest of our study.

4 Differential expression (DE)

4.1 The linear model

During this section, we model the DE of Hyperdiploid>50 samples relative to healthy ones. We use the limma package [5] and follow its pipeline, as described in [5] and [9].

First we compute RMA again, this time without the chips of bad quality.

$$Y_{gi} = n_g + h_i\beta_g + \epsilon_{gi}$$

where:

- Y_{gi} is the observed expression of g in the chip i
- n_g is the intercept, so in our case the expression of g in healthy individuals
- β_g is a coefficient that models the difference of expression with respect to the controls
- h_i is a dummy variable, equal to 0 if i is from a healthy individual and 1 otherwise.
- ϵ_{gi} is an error term. We assume that $\epsilon_{gi} \sim \mathcal{N}(0, \sigma_g^2)$, where the variance σ_g^2 depends on the gene g and is unknown

In particular, we have a linear model with the design matrix $X = [\vec{v} \ \vec{w}]$, where $\vec{v}$ is a $\mathbf{1}$ vector (a column vector of ones) and $\vec{w}$ a column vector with the h_i in order. This model is fitted for each gene using an empirical Bayes method. Note that in this case we do not need to consider a contrast matrix, as the comparison of interest is given in the design matrix.

4.2 Measuring differential expression, multiple test adjustment

Our strategy to rank DE is the following: first we determine the subset of DE genes. For this we perform hypothesis tests for each gene ($H_0 : \beta_g = 0, H_1 : \beta_g \neq 0$), and restrict our list of genes to those with low enough p -values. Then, we rank the genes in this subset by their log-fold change³, as we already have statistical evidence of DE.

Given the relatively low number of samples, the standard t-statistic is not an ideal option, as it is sensitive to genes with high variance. Therefore we employ the moderated-t statistic (mod-t), an adapted version of the t statistic described in [1]. In particular, for each gene g we have

$$t_{g,mod} = \frac{\hat{\beta}_g}{s'_g / \sqrt{v_g}}$$

where:

- $\hat{\beta}_g$ is our estimate for β_g
- v_g is a known value depending on the design matrix
- s'_g is the posterior variance of g , given as a weighted average of s_g (the residual standard deviation of the model) and s_0 (the prior mean variance of the genes).

We also want to adjust our p -values for the mod-t statistic. This is important: suppose we set 0.05 as a cut-off p -value. We are working with more than 10000 genes, so by pure chance we could get $\frac{5}{100} * 10000 = 500$ genes that are falsely reported as DE. Therefore, it is essential to modify our p -values and make them stricter.

For this, we use the Benjamini-Hochberg (BH) method; fix a threshold value $\alpha \in [0, 1]$ (in our case, $\alpha = 0.05$).

1. We sort the p -values by ascendant order (so that we have a list $\{p_1 \leq p_2 \leq \dots \leq p_n\}$).
2. Then, the adjusted p -values are $p'_x := \min\{\frac{n}{x} * p_x, p'_{x+1}\}$ for $x < n$, and $p'_n := p_n$.
3. Finally, we define the DE genes as those whose adjusted p -value p' have $p' < \alpha$.

³for a probe set g , the log-fold change is $\bar{a}^{(h)} - \bar{a}^{(n)}$, where $\bar{a}^{(h)}$ is the average of chip effects of g over Hyperdiploid>50 samples, and $\bar{a}^{(n)}$ the average of chip effects over healthy samples

A main feature of the BH method is the following: define the false discovery rate (FDR) as the expected value of

$$\frac{|\{g \in D | g \text{ is not DE}\}|}{|D|}$$

where D is the set of genes that our test determines to be DE⁴. Then, the BH method bounds the FDR at $FDR \leq \alpha$, as shown in [2].

In our study, we set a threshold of $\alpha = 0.05$ for the adjusted p -values. This gives us that our estimated subset of DE genes has 1357 elements. A table of the 50 genes with highest (absolute) log-fold change is included at the end of the report.

4.3 Visualisation

Our situation is represented in the following volcano plot, where we plot the adjusted p -values ($-\log_{10}(\text{adj}P)$)⁵ with the log-fold change (M) :

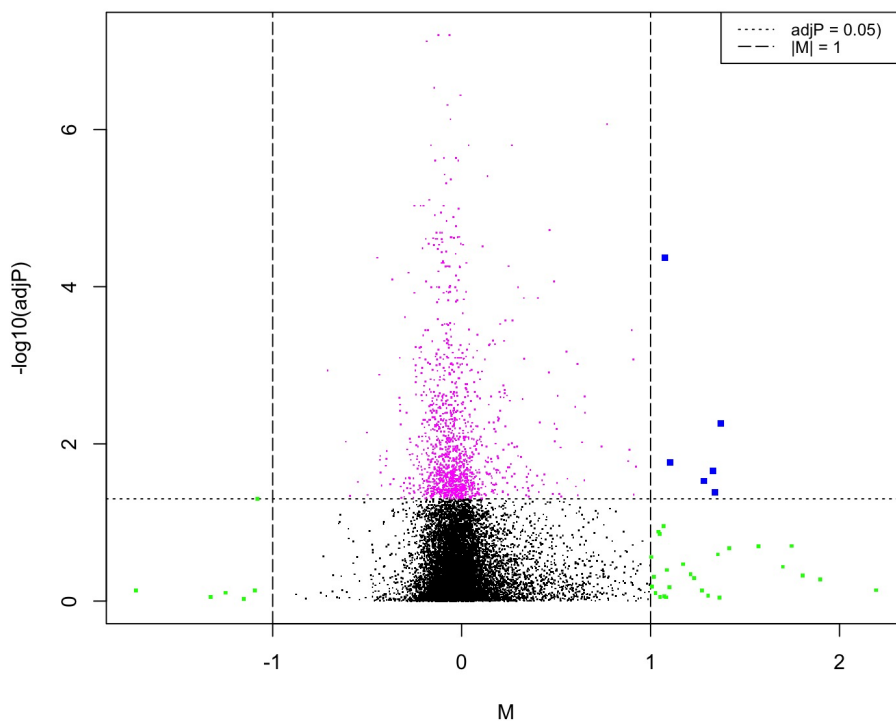


Figure 5: Volcano plot comparing adjusted p -values with log-fold change

in the plot, we partition the set of genes in four subsets:

1. In blue: genes with $|M| > 1$ and $\text{adj}P < 0.05$. These genes have a bigger difference in gene expression, and we can say that they are DE with statistical confidence. Thus, according to our study they should be the most relevant subgroup of genes. The genes are 1065_at, 1520_s.at, 1915_s.at, 306_s.at, 307_at, and 32207_at.
2. In pink: $|M| \leq 1$ and $\text{adj}P < 0.05$. Our study indicates statistical significance of DE for these genes. However, their biological relevance is not guaranteed.
3. In green: $|M| > 1$ and $\text{adj}P \geq 0.05$. They could be biologically relevant, but there is no strong statistical evidence (they could just have higher variability).
4. In black: $|M| \leq 1$ and $\text{adj}P \geq 0.05$. There is no statistical nor biological evidence of DE for these genes.

Note the low number of blue genes, and high amount of pink and black ones. From this, we hypothesise that the difference in gene expression between healthy individuals and the Hyperdiploid > 50 subgroup is subtle.

⁴if $|D| = 0$, we fix $FDR = 0$

⁵the adjusted p -values are transformed $-\log_{10}$ in order to improve the readability of the plot

5 Cluster analysis on the Hyperdiploid>50 subgroup

We perform sample clustering over the Hyperdiploid>50 subgroup in R ([6]).

5.1 Initial preparations

First, we filter out the control probes, as they are not useful for our analysis.

Then, we take the subset of the 1000 genes with biggest variance in expression value. This way we reduce noise, as the genes with low variance contribute less in the identification of subgroups.

It is worth noting that, if we do this process including the control probes, we notice that they rank unusually high in variance. In particular, the 10 most variable control probes are among the 500 most variable ones in the whole group of probe sets:

Rank (within control probes)	1	2	3	4	5	6	7	8	9	10
Overall rank	47	73	101	102	122	136	181	220	230	338

This could point to technical defects in our dataset, as control probes should rank low in variance by their definition. However, we checked for this already during quality control. A multidimensional scaling (MDS) plot of the samples does not seem to indicate batch effects either:

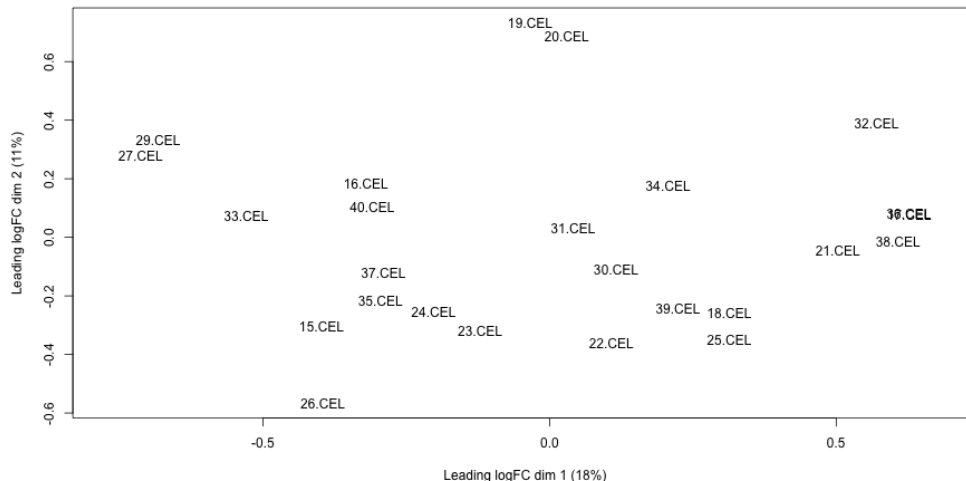


Figure 6: MDS plot of the Hyperdiploid>50 subgroup

5.2 Clustering

To compute the clusters, we employ hierarchical clustering. As outlined in [7], this method creates a binary tree of clusters according to the similarity of samples (so one expects resembling samples to be grouped closer in the tree). As a measure of similarity, we input a distance matrix of the samples.

There are three choices to make:

- The measure of distance: in this case, we use correlation-based distance. For two samples i, j we define it as $d(i, j) = 1 - \text{corr}(X_i, X_j)$, where X_i is the vector of gene expression values for i across the 1000 genes (same for X_j), and $\text{corr}(X_i, X_j)$ is the Pearson correlation coefficient between them.
- The method of clustering: we use Ward's method, as described in section 2.1 of [8]. Ward's method is an agglomerative algorithm, so at each cut (step) of the binary tree we merge two clusters. These are chosen so that the sum of the variances within each cluster is minimal.
- The number of clusters k : For hierarchical clustering, we fix the number of clusters

For the last point, we perform an average silhouette analysis in order to determine good choices for the number of clusters: given a set of clusters (in our case, the one we have in our

binary tree), this process measures for each one the mean dissimilarity of samples with their cluster compared to other clusters.

More specifically, fix a sample i and a clustering. We define $a(i)$ as the average dissimilarity between i and other samples in its cluster, and $b(i)$ as the lowest average dissimilarity between i and other samples in any cluster. Then, the silhouette width is defined as

$$s(i) = \frac{b(i) - a(i)}{\max\{b(i), a(i)\}}$$

Ideally, a good clustering should have a high average silhouette. In our case, the three best clusterings (or equivalently, choices of k) are:

N ^o of clusters (k)	2	3	4
Average Sil.	0.2124577	0.1893409	0.1803452

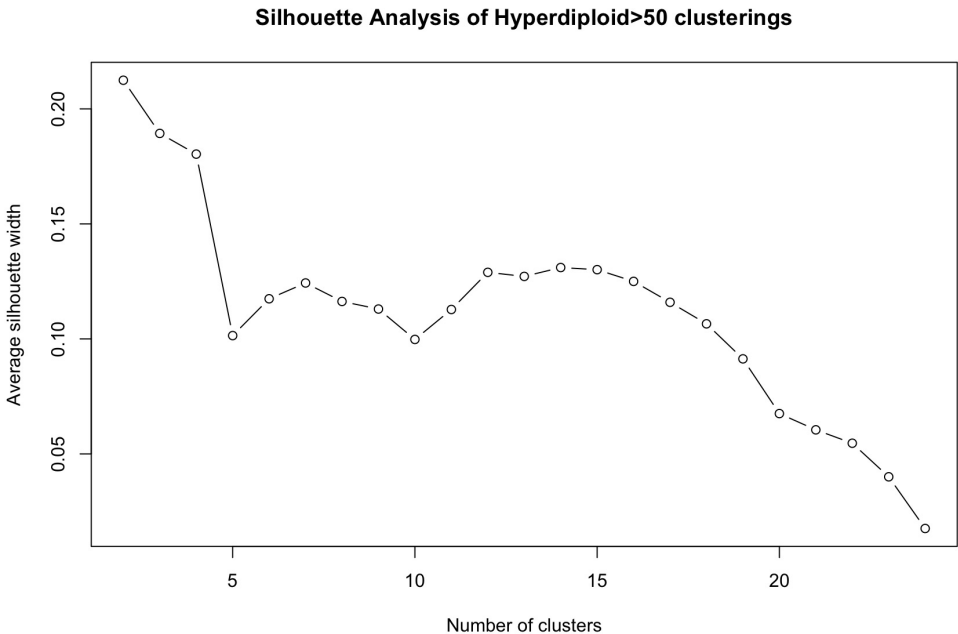


Figure 7: Plot of average silhouette width for different choices of k

We can visualise this in the form of silhouette plots:

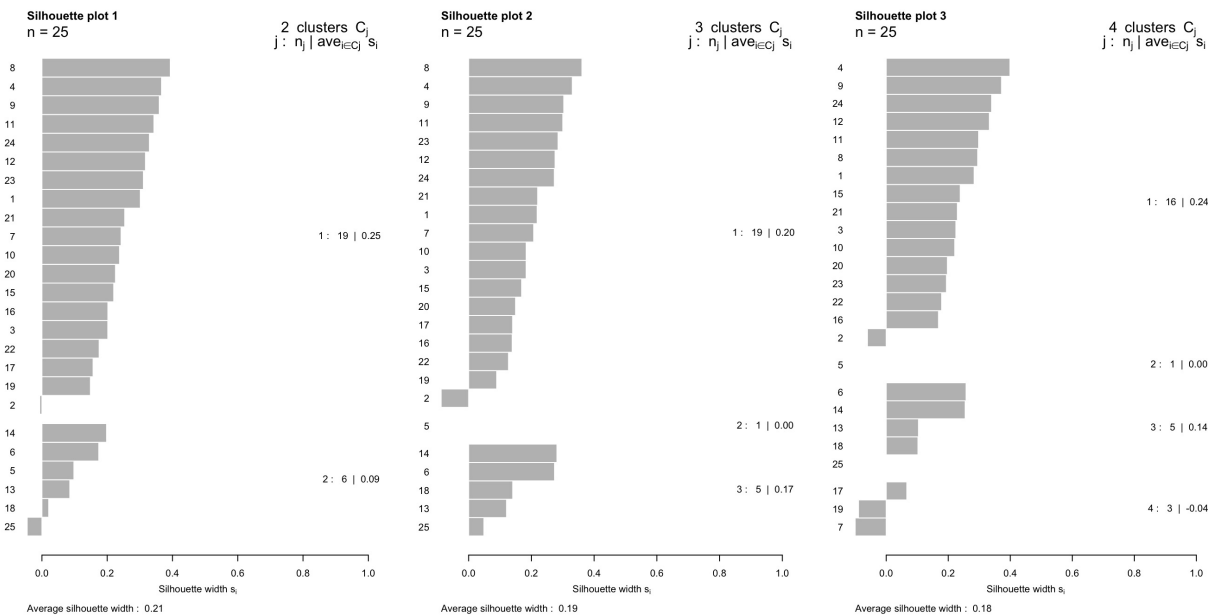


Figure 8: Silhouette plots for clusterings with 2,3 and 4 clusters

The clusterings have low average silhouette, which indicates a lack of significant structure.

We focus on the case with highest average silhouette (2 clusters). To visualise it, we plot a heat map:

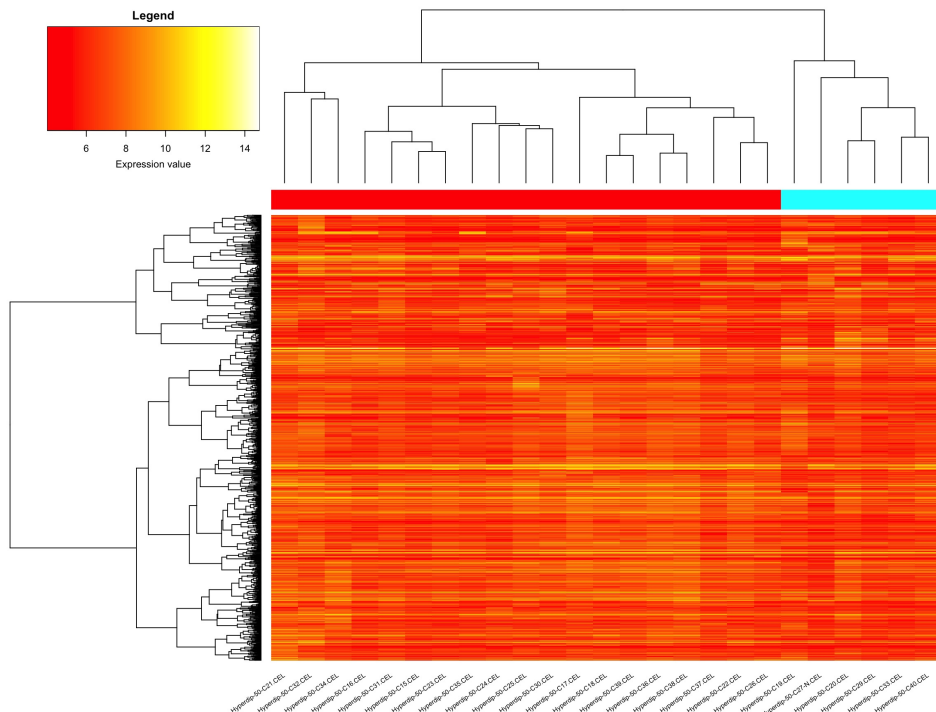


Figure 9: Heat map of the clustering with $k = 2$

Recall that we have a relatively high variance on the control probes, as well as a low average silhouette in our clusterings. In addition, we fail to infer an evident structure from the heat map. Therefore, we can not identify a subgroup in the Hyperdiploid>50 cases.

6 Conclusion

As a recapitulation, we have gone through the following procedure:

1. We start by normalising our data using RMA, in order to properly compare expression values across chips.
2. We perform a basic quality control routine, by checking for artifacts in the pseudo-images and computing NUSE.
3. We study DE: first, we define a linear model with our (implicit) comparison of interest. Then, we fit the model for each gene with an empirical Bayes procedure, and adjust for multiple testing. Finally, we determine the DE genes as those with low enough (adjusted) p -values for the moderated t statistic.
4. We perform a cluster analysis over the Hyperdiploid>50 samples. For this, we first set a measure of distance for samples (correlation based), and our method of clustering (Ward's method, which is hierarchical). Then, we identify 3 candidate choices for the number of clusters with a silhouette analysis. From this, we take the one with highest average silhouette. Lastly, we visualised the clustering with a heat map to look for visual hints of structure.

Our analysis estimates that a relatively small proportion of the genes (1357 out of 12558) are DE between patients of ALL Hyperdiploid>50 and healthy individuals. From these genes, only 6 have absolute log fold change greater than 1. Therefore, we infer that differential expression is rather subtle, and the disease may have a stronger link with some of the 6 genes.

Concerning the cluster analysis, we found some potential subgroups in our sample. However, we find indicators that argue against them reflecting meaningful structure:

- A low average silhouette for all clusterings
- A considerable amount of control probes ranking high in variance between samples

Therefore, we do not present our clusterings as subgroups.

References

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- [2] Yoav Benjamini and Yosef Hochberg. “Controlling the false discovery rate: a practical and powerful approach to multiple testing”. In: *Journal of the Royal Statistical Society: Series B (Methodological)* 57.1 (1995).
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- [6] Martin Maechler et al. *cluster: Cluster Analysis Basics and Extensions*. R package version 2.1.8.1 — For new features, see the ‘NEWS’ and the ‘Changelog’ file in the package source). 2025. URL: <https://CRAN.R-project.org/package=cluster>.
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- [9] ”Gordon K. Smyth et al. *Linear Models for Microarray Data User’s Guide*. ”The Walter and Eliza Hall Institute of Medical Research”, 2007.
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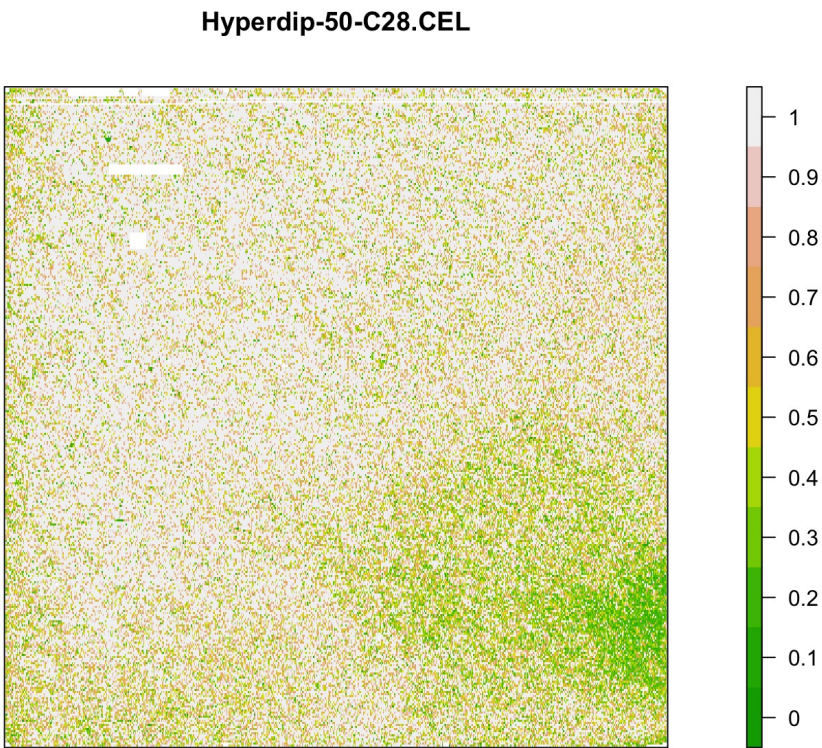
A Top 50 most DE genes

	Log FC	Avrg. Expression	Mod. <i>t</i>	Raw <i>p</i> -Value	Adj. <i>p</i> -Value	<i>B</i> stat.
38604_at	2.19	7.62	7.52	0.00	0.00	11.61
38968_at	1.90	7.79	8.86	0.00	0.00	15.81
36620_at	1.80	7.54	8.17	0.00	0.00	13.92
35939_s_at	1.75	8.02	6.02	0.00	0.00	6.83
39878_at	-1.72	6.39	-5.01	0.00	0.00	3.40
37014_at	1.70	7.76	4.80	0.00	0.00	2.78
1916_s_at	1.57	7.32	4.79	0.00	0.00	2.81
35816_at	1.42	7.33	5.99	0.00	0.00	6.71
1915_s_at	1.37	8.35	4.08	0.00	0.01	0.61
41097_at	1.37	8.51	4.75	0.00	0.00	2.66
36021_at	1.36	8.05	5.62	0.00	0.00	5.47
32207_at	1.34	7.80	7.42	0.00	0.00	11.73
1520_s_at	1.33	7.60	6.32	0.00	0.00	7.87
41266_at	-1.33	7.29	-3.51	0.00	0.02	-0.84
39781_at	1.30	6.52	6.46	0.00	0.00	8.17
307_at	1.28	7.20	5.14	0.00	0.00	3.94
39402_at	1.27	7.25	6.58	0.00	0.00	8.71
37677_at	1.23	8.81	6.45	0.00	0.00	8.16
37398_at	1.21	6.72	4.11	0.00	0.01	0.71
35940_at	1.17	5.96	4.88	0.00	0.00	3.10
578_at	-1.15	5.81	-6.23	0.00	0.00	7.39
306_s_at	1.10	9.14	7.76	0.00	0.00	12.96
37350_at	1.10	6.88	8.21	0.00	0.00	14.31
39069_at	-1.10	6.82	-6.39	0.00	0.00	8.06
39065_s_at	1.09	6.04	5.99	0.00	0.00	6.66
41139_at	1.08	7.21	5.82	0.00	0.00	6.20
33410_at	-1.08	5.84	-4.17	0.00	0.00	0.96
1065_at	1.08	7.70	4.11	0.00	0.01	0.76
40607_at	1.07	7.08	4.27	0.00	0.00	1.18
1389_at	1.07	8.62	4.03	0.00	0.01	0.52
41470_at	1.05	5.20	3.66	0.00	0.01	-0.55
32542_at	1.05	6.55	4.43	0.00	0.00	1.78
31863_at	1.04	6.23	8.84	0.00	0.00	16.28
40019_at	1.03	6.34	3.46	0.00	0.02	-1.03
38527_at	1.02	9.65	7.47	0.00	0.00	11.83
38518_at	1.01	6.43	8.65	0.00	0.00	15.70
36629_at	1.00	8.90	4.15	0.00	0.00	0.63
38458_at	1.00	6.75	4.24	0.00	0.00	1.15
38738_at	0.99	7.68	7.92	0.00	0.00	13.49
37326_at	0.99	7.39	4.99	0.00	0.00	3.56
37539_at	0.99	6.21	6.67	0.00	0.00	8.98
41724_at	0.99	8.69	6.17	0.00	0.00	7.42
38459_g_at	0.98	6.50	4.43	0.00	0.00	1.69
39345_at	0.97	7.00	5.96	0.00	0.00	6.66
36108_at	0.96	6.52	3.07	0.00	0.04	-1.97
40771_at	0.95	9.80	5.52	0.00	0.00	5.08
36946_at	0.95	6.66	5.30	0.00	0.00	4.43
33641_g_at	0.95	6.13	6.15	0.00	0.00	7.22
38408_at	0.94	6.33	4.71	0.00	0.00	2.62
38065_at	0.94	8.50	4.35	0.00	0.00	1.31

Table 1: Top 50 DE genes

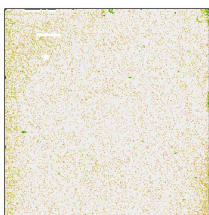
B List of chip pseudo-images

Please see the two following pages for a complete list of the pseudo-images of the (weight of the) chips. They follow the same legend as the one displayed in Figure 1, which is printed again for convenience:

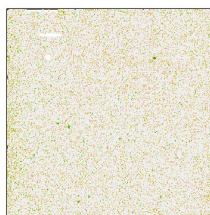




Hyperdip-50-C39.CEL



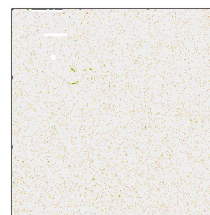
Hyperdip-50-C40.CEL



Normal-#1.CEL



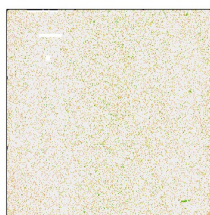
Normal-#2.CEL



Normal-#3.CEL



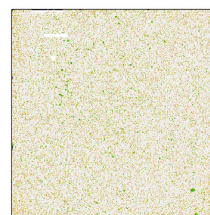
Normal-C1-N.CEL



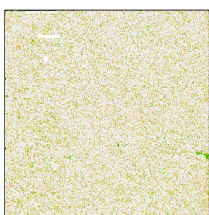
Normal-C10.CEL



Normal-C11-N.CEL



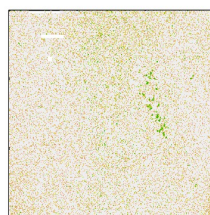
Normal-C12.CEL



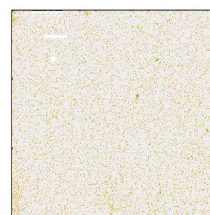
Normal-C2-N.CEL



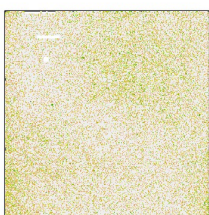
Normal-C3-N.CEL



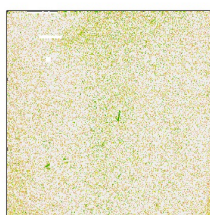
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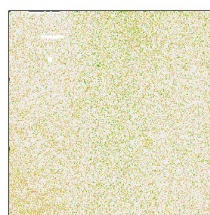
Normal-C5.CEL



Normal-C6.CEL



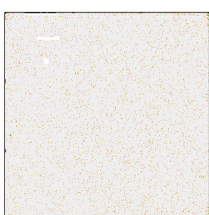
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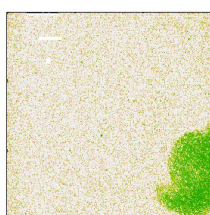
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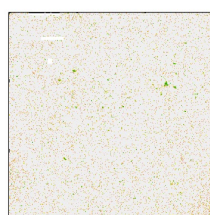
Normal-C9.CEL



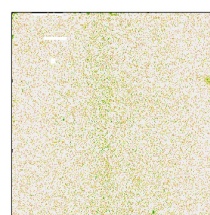
Normal-R1.CEL



Normal-R2-N.CEL



Normal-R3.CEL



Normal-R4.CEL

